

i) w^* & μ^* are feasible for primal & dual respectively.

$$ii) L(w^*, \mu^*) = \min_w L(w, \mu^*)$$

The Dual function :

$$q(\mu) = \inf_{w, b} \left\{ \frac{1}{2} w^T w + \sum_{i=1}^n \mu_i (1 - y_i (w^T x_i + b)) \right\}$$

Observe the term $\sum_i \mu_i y_i b_i$, which can drive the inf. to $-\infty$.

$\therefore$ Solve a constrained version, by forcing $\sum_i \mu_i y_i = 0$

To solve the problem, substitute

$$w^* = \sum_{i \in S} \mu_i y_i x_i \quad b^* = ? \quad (\text{H.W.})$$

$$q(\mu) = \sum_{i=1}^n \mu_i - \frac{1}{2} \sum_i \sum_j \mu_i y_i \mu_j y_j x_i^T x_j$$

In the dual problem, the data x only appears as inner products.

Dual optimization problem :

$$\begin{aligned} \max_{\mu} \quad & \sum_i \mu_i - \frac{1}{2} \sum_i \sum_j \mu_i \mu_j y_i y_j x_i^T x_j \\ \text{s.t.} \quad & \mu_i \geq 0 \\ & \sum_i y_i \mu_i = 0 \end{aligned}$$

The above problem is quadratic in μ with linear constraints.

use standard qp solvers
for instance Sequential Min. opti
(SMO)

libsvm.

once μ^* are obtained,
find

$$w^* = \sum_s y_i \mu_i^* x_i$$

Support vectors are those datapoints
for which $\mu^* > 0$

from one of the KKT cond.

$$\mu_i^* [1 - y_i (w^T x_i + b)] = 0$$

for support vectors,

$$y_i (w^T x_i + b) = 1.$$

SVM for not linear-separable case

$$\Rightarrow \nexists w \text{ s.t. } y_i (w^T x_i + b) > 1 \quad \forall i$$

Introduce another variable into optimization that quantifies the extent of mis-classification.

The original problem:

$$\min_w \quad \frac{1}{2} w^T w$$

$$\text{s.t.} \quad y_i (w^T x_i + b) > 1 - \xi_i$$

↪ slack variable

unconstrained slack may lead to a trivial solution (for e.g. $w=0$)

∴ solve a constrained version,

$$\min_{w, \xi} \quad \frac{1}{2} w^T w + C \sum_{i=1}^n \xi_i$$

$$\text{s.t.} \quad y_i (w^T x_i + b) \geq 1 - \xi_i \quad \forall i$$
$$\xi_i \geq 0.$$

$$L(w, b, \xi, \mu, d) = \frac{1}{2} w^T w + C \sum_{i=1}^n \xi_i +$$
$$\sum_i \mu_i \left(1 - \xi_i - y_i (w^T x_i + b) \right) -$$
$$\sum d_i \xi_i$$

KKT for the above problem :

$$\text{i)} \quad \nabla_w L = 0 \Rightarrow w^* = \sum_i \mu_i^* y_i x_i$$

$$\text{ii)} \quad \nabla_b L = 0 \Rightarrow \sum_i \mu_i^* y_i = 0$$

$$\text{iii)} \quad \nabla_{\xi} L = 0 \Rightarrow \mu_i^* + d_i^* = C$$

$$\text{iv)} \quad \mu_i \left(1 - \xi_i - y_i (w^T x_i + b) \right) = 0 \quad \forall i$$
$$d_i \xi_i = 0$$

Formulating the Dual.

$$Q(M, d) = \inf_{w, b, \xi} L(w, b, \xi, M, d)$$

$$\sum y_i b = 0$$

Observe that L has a term

$$\sum_i (C - \mu_i - d_i) \xi_i, \text{ that needs}$$

to be made zero to avoid the trivial soln.

$$\Rightarrow \mu_i + d_i = C.$$

Dual problem :

$$\max_{\mu, d} \sum_i \mu_i - \frac{1}{2} \sum_i \sum_j y_i y_j \mu_i \mu_j x_i^T x_j$$

$$\text{s.t. } \mu_i \geq 0$$

$$d_i \geq 0 \text{ and } C = \mu_i + d_i$$

To take out d_i , modify the constraint as $0 \leq \mu_i \leq C$.

$$\max_{\mu} \sum_i \mu_i - \frac{1}{2} \sum_i \sum_j y_i y_j \mu_i \mu_j x_i^T x_j$$

$$\text{s.t. } \mu_i \geq 0,$$

$$0 \leq \mu_i \leq C.$$

SVM with the kernel functions.

Define a function $\phi: \mathbb{R}^d \rightarrow \mathbb{R}^{d'}$

typically $d' \gg d$

with this, the dual fn. to be solved will be

$$\begin{aligned} \max_{\mu} \quad & \sum \mu_i - \frac{1}{2} \sum_i \sum_j \mu_i \mu_j y_i y_j \phi(x_i)^T \phi(x_j) \\ \text{s.t.} \quad & 0 \leq \mu_i \leq C \end{aligned}$$

suppose $\exists$ a function k :

$$k: \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R} \quad \text{s.t.}$$

$$k(x_i, x_j) = \phi(x_i)^T \phi(x_j) \quad \forall i, j.$$

↳ kernel function.

The SVM dual problem will be now

$$\sum \mu_i - \frac{1}{2} \sum_i \sum_j \mu_i \mu_j y_i y_j k(x_i, x_j)$$

Mercer's theorem : suppose $\bar{K}$ is an $n \times n$ matrix with $\bar{K}_{ij} = k(x_i, x_j)$
if $\bar{K}$ is PSD $\forall D$, then k is a valid kernel. (one that corresponds to a proj. ϕ)

Examples of valid kernels.

i) poly. kernel : $k_p(x_1, x_2) = (1 + x_1^T x_2)^p$

ii) Sigmoid kernel : $k_s(x_1, x_2)$
 $= \tanh(a x_1^T x_2 + b)$

$$\begin{aligned} g(\phi(x)) &= W^T \phi(x) \\ &= \left[\sum_s \mu_i^* y_i \phi(x_i) \right]^T \phi(x) \\ &= \sum_s \mu_i y_i \tanh(a x_i^T x + b) \end{aligned}$$

iii) Gaussian kernel :

$$k_a(x_1, x_2) = e^{-\frac{\|x_1 - x_2\|^2}{\sigma^2}}$$

SVM as Regularized ERM

$$\min_{w, b, \xi} \frac{1}{2} w^T w + C \sum \xi_i$$

$$\text{s.t. } y_i (w^T x_i + b) \geq 1 - \xi_i$$

$$\xi_i \geq 0$$

Given any w & b , ξ_i has to satisfy,

$$\xi_i \geq \max \left(0, 1 - y_i (w^T x_i + b) \right)$$

$$\min_{w, b} \frac{1}{2} w^T w + C \sum_{i=1}^n \max \left(0, 1 - y_i (w^T x_i + b) \right)$$

$$\Omega(w) + C \sum L(h(x_i), y_i)$$

$$L(h(x), y) = \max \left(0, 1 - y \cdot h(x) \right)$$

↪ hinge-loss.